

Source: Analysis III (MA 321), Lecture notes and course material.

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Area: Distribution Theory

Type: Problem Solving

Status: Personal solution with detailed derivation.

Date: February 2026

Analysis III – Assignment 1: Distributional Derivative of $\log |x|$

This document contains a complete solution to Problem 1 of Assignment I from the Analysis III course at IISc. The problem illustrates how a locally integrable function defines a distribution and how its distributional derivative is computed.

Problem 0.1 (Problem 1). *Show that $f = \log |x|$ is locally integrable and hence defines a distribution. Calculate $\frac{d}{dx} \log |x|$.*

Solution. **Local integrability.** The function $f = \log |x|$ diverges at $x = 0$. To prove local integrability, it suffices to examine a compact set containing the singular point. Let $K \subset \subset \mathbb{R}^n$ be compact. Then $K \subset B(0, R)$ for some $R > 0$. Hence

$$\int_K |\log |x|| dx \leq \int_{B(0, R)} |\log |x|| dx.$$

For $\epsilon > 0$, split the integral:

$$\int_{B(0, R)} |\log |x|| dx = \int_{B(0, \epsilon)} |\log |x|| dx + \int_{\epsilon \leq |x| \leq R} |\log |x|| dx.$$

The second integral is finite because $\log |x|$ is bounded on the annulus $\epsilon \leq |x| \leq R$. For the first, use polar coordinates:

$$\int_{B(0, \epsilon)} |\log |x|| dx = \omega_{n-1} \int_0^\epsilon |\log r| r^{n-1} dr,$$

where ω_{n-1} is the surface area of the unit sphere. The integral is finite since

$$\int_0^\epsilon |\log r| r^{n-1} dr < \infty,$$

which follows from the fact that r^{n-1} dominates the logarithmic singularity near $r = 0$. Hence

$$\int_{B(0, \epsilon)} |\log |x|| dx < \infty.$$

Therefore $f \in L^1_{\text{loc}}(\mathbb{R}^n)$.

Definition as a distribution. For any test function $\phi \in \mathcal{D}(\mathbb{R})$, define

$$\langle T_f, \phi \rangle = \int_{\mathbb{R}} \log |x| \phi(x) dx.$$

The integral converges absolutely because ϕ has compact support and f is locally integrable. We verify the two defining properties of a distribution:

- *Linearity:* For any $\phi, \psi \in \mathcal{D}(\mathbb{R})$ and $a, b \in \mathbb{R}$,

$$\langle T_f, a\phi + b\psi \rangle = a\langle T_f, \phi \rangle + b\langle T_f, \psi \rangle.$$

- *Continuity:* Let $K \subset \mathbb{R}$ be compact and $\{\phi_j\} \subset \mathcal{D}(\mathbb{R})$ with $\text{supp } \phi_j \subset K$ such that $\phi_j \rightarrow 0$ in $\mathcal{D}(\mathbb{R})$. Then

$$|\langle T_f, \phi_j \rangle| \leq \int_K |\log |x|| |\phi_j(x)| dx \leq \sup_{x \in K} |\phi_j(x)| \int_K |\log |x|| dx.$$

Since $\int_K |\log |x|| dx < \infty$ and $\sup_K |\phi_j| \rightarrow 0$, we have $\langle T_f, \phi_j \rangle \rightarrow 0$.

Thus the functional T_f defines a distribution associated with the locally integrable function $\log |x|$. By abuse of notation, this distribution is also denoted by $\log |x|$.

Distributional derivative. By definition,

$$\left\langle \frac{d}{dx} \log |x|, \phi \right\rangle = - \langle \log |x|, \phi' \rangle = - \int_{\mathbb{R}} \log |x| \phi'(x) dx \quad \forall \phi \in \mathcal{D}(\mathbb{R}).$$

Interpret the integral as the improper limit:

$$- \int_{-\infty}^{\infty} \log |x| \phi'(x) dx = - \lim_{\varepsilon \rightarrow 0^+} \left(\int_{-\infty}^{-\varepsilon} + \int_{\varepsilon}^{\infty} \right) \log |x| \phi'(x) dx.$$

On each interval, $\log |x|$ is smooth. Integrate by parts:

$$\int_{\varepsilon}^{\infty} \log x \phi'(x) dx = - \log \varepsilon \phi(\varepsilon) - \int_{\varepsilon}^{\infty} \frac{\phi(x)}{x} dx,$$

and

$$\int_{-\infty}^{-\varepsilon} \log |x| \phi'(x) dx = \log \varepsilon \phi(-\varepsilon) - \int_{-\infty}^{-\varepsilon} \frac{\phi(x)}{x} dx.$$

Adding and multiplying by -1 :

$$- \left(\int_{\varepsilon}^{\infty} + \int_{-\infty}^{-\varepsilon} \right) \log |x| \phi'(x) dx = \log \varepsilon (\phi(\varepsilon) - \phi(-\varepsilon)) + \int_{\varepsilon}^{\infty} \frac{\phi(x)}{x} dx + \int_{-\infty}^{-\varepsilon} \frac{\phi(x)}{x} dx.$$

As $\varepsilon \rightarrow 0^+$, the first term vanishes because $\phi(\varepsilon) - \phi(-\varepsilon) = O(\varepsilon)$. Hence

$$\left\langle \frac{d}{dx} \log |x|, \phi \right\rangle = \text{pv} \int_{-\infty}^{\infty} \frac{\phi(x)}{x} dx.$$

Therefore,

$$\boxed{\frac{d}{dx} \log |x| = \text{pv} \frac{1}{x}}$$

in the sense of distributions.

Structural Interpretation. Although $\log |x|$ is singular at the origin, its singularity is sufficiently mild to remain locally integrable. Distribution theory therefore allows us to treat $\log |x|$ as a generalized function.

The appearance of the principal value distribution reflects the fact that differentiation amplifies the singularity from a locally integrable logarithm to the non-integrable function $1/x$. The principal value provides the natural extension of differentiation beyond classical functions and illustrates one of the central ideas of weak analysis: singular objects can still possess well-defined derivatives in the distributional sense.